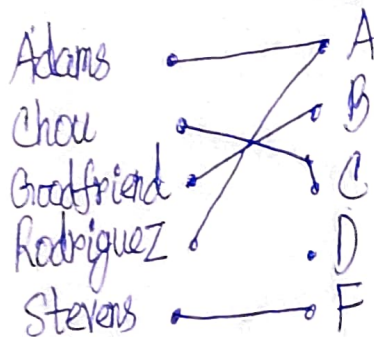


Solution: $F \cap R = \{ \min(0.6, 0.4) \text{ Alice}, \min(0.9, 0.8) \text{ Brian}, \min(0.4, 0.2) \text{ Fred}, \min(0.1, 0.9) \text{ Oscar}, \min(0.5, 0.7) \text{ Rita} \}$
 $= \{ 0.4 \text{ Alice}, 0.8 \text{ Brian}, 0.2 \text{ Fred}, 0.1 \text{ Oscar}, 0.5 \text{ Rita} \} \square$

2.3 Functions

2.3.1 Introduction

In many instances, we assign to each element of a set a particular element of a second set (which may be the same as the first). For example, suppose that each student in a discrete mathematics class is assigned a letter grade from the set $\{A, B, C, D, F\}$. And suppose that the grades are A for Adams, C for Chou, B for Goodfriend, A for Rodriguez and F for Stevens.



This assignment is an example of a function

Figure 1: Assignment of grades in a discrete mathematics class

Definition 1: Let A and B be nonempty sets. A **function** f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f: A \rightarrow B$.

Remark: Functions are sometimes also called **mappings** or **transformations**.

- A function $f: A \rightarrow B$ can also be defined in terms of a relation from A to B .
- A relation from A to B that contains one, and only one, ordered pair (a, b) for every element a of A , defines a function f from A to B . This function is defined by the assignment $f(a) = b$, where (a, b) is the unique ordered pair in the relation that has a as its first element.

~~map~~